MSG No. 85.65 P_C4/n [Type IV, tetragonal]

Table 1: Wyckoff site: 2a, site symmetry: 4/m'..

No.	position	mapping
1	$\left[\frac{3}{4}, \frac{3}{4}, 0\right]$	[1,2,3,4,13,14,15,16]
2	$\left[\frac{1}{4}, \frac{1}{4}, 0\right]$	[5,6,7,8,9,10,11,12]

Table 2: Wyckoff site: 2b, site symmetry: 4/m'..

No.	position	mapping
1	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{2}\right]$	[1,2,3,4,13,14,15,16]
2	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{2}\right]$	[5,6,7,8,9,10,11,12]

Table 3: Wyckoff site: 2c, site symmetry: 4'/m'..

No.	position	mapping
1	$\left[\frac{3}{4}, \frac{1}{4}, 0\right]$	[1,4,6,7,10,11,13,16]
2	$\left[\frac{1}{4}, \frac{3}{4}, 0\right]$	[2,3,5,8,9,12,14,15]

Table 4: Wyckoff site: 2d, site symmetry: 4'/m'..

No.	position	mapping
1	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}\right]$	[1,4,6,7,10,11,13,16]
2	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{2}\right]$	[2,3,5,8,9,12,14,15]

Table 5: Wyckoff site: 4e, site symmetry: 2'/m'..

No.	position	mapping
1	$\left[\frac{1}{2}, 0, 0\right]$	[1,5,12,16]
2	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	[2,6,11,15]
3	$[0, 0, 0]$	[3,7,10,14]
4	$\left[0, \frac{1}{2}, 0\right]$	[4,8,9,13]

Table 6: Wyckoff site: 4f, site symmetry: 2'/m'..

No.	position	mapping
1	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	[1,5,12,16]
2	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	[2,6,11,15]
3	$\left[0, 0, \frac{1}{2}\right]$	[3,7,10,14]
4	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	[4,8,9,13]

Table 7: Wyckoff site: `4g`, site symmetry: `4..`

No.	position	mapping
1	$\left[\frac{3}{4}, \frac{3}{4}, z\right]$	`[1,2,3,4]`
2	$\left[\frac{1}{4}, \frac{1}{4}, -z\right]$	`[5,6,7,8]`
3	$\left[\frac{1}{4}, \frac{1}{4}, z\right]$	`[9,10,11,12]`
4	$\left[\frac{3}{4}, \frac{3}{4}, -z\right]$	`[13,14,15,16]`

Table 8: Wyckoff site: `4h`, site symmetry: `4'..`

No.	position	mapping
1	$\left[\frac{3}{4}, \frac{1}{4}, z\right]$	`[1,4,10,11]`
2	$\left[\frac{1}{4}, \frac{3}{4}, z\right]$	`[2,3,9,12]`
3	$\left[\frac{1}{4}, \frac{3}{4}, -z\right]$	`[5,8,14,15]`
4	$\left[\frac{3}{4}, \frac{1}{4}, -z\right]$	`[6,7,13,16]`

Table 9: Wyckoff site: `8i`, site symmetry: `2'..`

No.	position	mapping
1	$\left[\frac{1}{2}, 0, z\right]$	`[1,12]`
2	$\left[\frac{1}{2}, \frac{1}{2}, z\right]$	`[2,11]`
3	$[0, 0, z]$	`[3,10]`
4	$\left[0, \frac{1}{2}, z\right]$	`[4,9]`
5	$\left[\frac{1}{2}, 0, -z\right]$	`[5,16]`
6	$\left[\frac{1}{2}, \frac{1}{2}, -z\right]$	`[6,15]`
7	$[0, 0, -z]$	`[7,14]`
8	$\left[0, \frac{1}{2}, -z\right]$	`[8,13]`

Table 10: Wyckoff site: `8j`, site symmetry: `m'..`

No.	position	mapping
1	$[x, y, 0]$	`[1,16]`
2	$\left[\frac{1}{2} - y, x, 0\right]$	`[2,15]`
3	$\left[y, \frac{1}{2} - x, 0\right]$	`[3,14]`
4	$\left[\frac{1}{2} - x, \frac{1}{2} - y, 0\right]$	`[4,13]`
5	$[-x, -y, 0]$	`[5,12]`
6	$\left[y + \frac{1}{2}, -x, 0\right]$	`[6,11]`
7	$\left[-y, x + \frac{1}{2}, 0\right]$	`[7,10]`
8	$\left[x + \frac{1}{2}, y + \frac{1}{2}, 0\right]$	`[8,9]`

Table 11: Wyckoff site: `8k`, site symmetry: `m'..`

No.	position	mapping
1	$\left[x, y, \frac{1}{2}\right]$	`[1,16]`
2	$\left[\frac{1}{2} - y, x, \frac{1}{2}\right]$	`[2,15]`

continued ...

Table 11

No.	position	mapping
3	$\left[y,\ \frac{1}{2}-x,\ \frac{1}{2}\right]$	[3,14]
4	$\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ \frac{1}{2}\right]$	[4,13]
5	$\left[-x,\ -y,\ \frac{1}{2}\right]$	[5,12]
6	$\left[y+\frac{1}{2},\ -x,\ \frac{1}{2}\right]$	[6,11]
7	$\left[-y,\ x+\frac{1}{2},\ \frac{1}{2}\right]$	[7,10]
8	$\left[x+\frac{1}{2},\ y+\frac{1}{2},\ \frac{1}{2}\right]$	[8,9]

Table 12: Wyckoff site: 16l, site symmetry: 1

No.	position	mapping
1	$[x,\ y,\ z]$	[1]
2	$\left[\frac{1}{2}-y,\ x,\ z\right]$	[2]
3	$\left[y,\ \frac{1}{2}-x,\ z\right]$	[3]
4	$\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ z\right]$	[4]
5	$[-x,\ -y,\ -z]$	[5]
6	$\left[y+\frac{1}{2},\ -x,\ -z\right]$	[6]
7	$\left[-y,\ x+\frac{1}{2},\ -z\right]$	[7]
8	$\left[x+\frac{1}{2},\ y+\frac{1}{2},\ -z\right]$	[8]
9	$\left[x+\frac{1}{2},\ y+\frac{1}{2},\ z\right]$	[9]
10	$\left[-y,\ x+\frac{1}{2},\ z\right]$	[10]
11	$\left[y+\frac{1}{2},\ -x,\ z\right]$	[11]
12	$[-x,\ -y,\ z]$	[12]
13	$\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ -z\right]$	[13]
14	$\left[y,\ \frac{1}{2}-x,\ -z\right]$	[14]
15	$\left[\frac{1}{2}-y,\ x,\ -z\right]$	[15]
16	$[x,\ y,\ -z]$	[16]